

CLAUDE

CLAUDE.md

INHERITED FROM constitution/CLAUDE.md

All rules in constitution/CLAUDE.md (and the constitution/Constitution.md it references) apply unconditionally. Project-specific rules below extend them — they do NOT weaken any universal clause. When this file disagrees with the constitution submodule, the constitution wins.

Propagated clauses (Helix Universal Constitution): §11.4.10 (credentials handling), §11.4.24 (build-resource stats tracking), §11.4.30 (gitignore discipline), §11.4.44 (document revision headers), §11.4.65 (universal markdown export), §11.4.66 (interactive clarification), §11.4.75 (mechanical enforcement), §11.4.78 (CodeGraph integration), §11.4.85 (stress + chaos test mandate), §11.4.109 (anti-forgetting enforcement), §11.4.113 (no force-push), §11.4.125 (code-review-agent gate), §11.4.142 (universal code-review — every change reviewed, always).

@constitution/CLAUDE.md

This file provides guidance to Claude Code (claude.ai/code) when working with code in this repository.

For deeper reference (technology stack, per-test-file mapping, full gotchas), see AGENTS.md.

Critical Constraints

- **Multi-track identity: this project is Track 11** in the operator's multi-track development fleet (permanently adopted 2026-08-08). Trunk (main) work always labels T1 per the constitution's own hard-

coded TRUNK RULE (never overridden); non-trunk (feature/product/flavor) work from this checkout labels T11 via scripts/multitrack/track_branch_label.sh (a boba-owned wrapper, never a copy, around the shared constitution labeler). See docs/MULTITRACK.md for the full rationale. Execution is single-track until the operator asks for parallel dispatch.

- **QA discovery-channel ledger (§11.4.238):** automated HelixQA coverage MUST be the discovery layer; manual QA / operator reports / agent-found defects are confirmation-only, and any defect found out-of-band is itself a coverage-escape release blocker, not merely a bug to fix. Every such finding gets a coverage-escape-audit entry (why the automated regime missed it) + a new or strengthened automated check with its own RED-capturing-the-escape evidence. See docs/QA_DISCOVERY_LEDGER.md for the schema, the real entries recorded so far, and the tracked discovery-channel split. scripts/pre_build_verification.sh invariant 17 is the closed-loop example: it now runs both `workable-items validate` AND `workable-items diff` because a real DB↔Markdown divergence (BOB-008) was found by an agent, not by this gate, before the fix.
- **TDD is MANDATORY for all bug fixes and features:**
 - Write failing test first (RED)
 - Watch it fail
 - Write minimal code to pass (GREEN)
 - Verify tests pass
 - Then commit
- **Anti-Bluff Verification (CONST-XII)** — Tests and challenges MUST prove the feature works for the END USER. A green run is a contract with the user. Specifically:
 - Assert on user-observable outcomes (DB rows, file content, response body fields, container state, DOM text, rendered HTML attributes, browser console errors) — NOT just status codes / “no error”.
 - Each new test must fail against a no-op stub of the feature it tests. If it doesn’t, it’s a bluff and must be strengthened or deleted.
 - Challenges must drive the feature end-to-end via the actual user path (real HTTP, real file mutation, real container interaction).
 - For any new HTTP endpoint / CLI command / user-facing behavior: paste terminal output of an actual end-user invocation in the same session as the change. No self-

certification words (“verified”, “tested”, “working”, “complete”, “fixed”, “passing”) without that pasted evidence.

- Flaky tests are bluffs; they must be hardened or deleted.
- Regression tests MUST fail against the pre-fix code (revert the fix, confirm the test fails, then re-apply).
- No hardcoded localhost / 127.0.0.1 for client-facing URLs. Any URL, API base, CORS origin, or service address returned to a browser MUST derive from the request’s Host header, window.location, or an explicit PUBLIC_HOST env var.
- See CONSTITUTION.md § XII for the full rule. Apply universally — every submodule and sub-project inherits.

- **Pick the right restart level** (verified 2026-04-19 against the real docker-compose.yml mount strategy; wrapper subcommands added 2026-08-08 per Hard Stop #3 — container orchestration is owned exclusively by the project’s own binary/orchestrator, start.sh — see docs/GOVERNANCE_AUDIT_2026-08-07.md GA-27). Operators MUST use the start.sh subcommand for each level — never type raw podman/docker commands directly:

- **Python source in download-proxy/src/ (including merge_service/*.py)** — bind-mounted via ./download-proxy:/config/download-proxy. Run ./start.sh --reload-python. *(Under the hood: clears __pycache__ inside the qbittorrent-proxy container via podman|docker exec qbittorrent-proxy find /config/download-proxy -name __pycache__ -type d -exec rm -rf {} +, then podman|docker restart qbittorrent-proxy — auto-detecting the runtime exactly like the rest of start.sh.)*
- **Plugin files in plugins/** — also bind-mounted through ./config:/config. After editing, run ./install-plugin.sh FIRST (copies plugins/X.py into config/qBittorrent/nova3/engines/ — that path IS the host side of the mount), THEN ./start.sh --reload-plugins to pick them up. A direct edit to config/qBittorrent/nova3/engines/X.py works for a one-shot try but will be clobbered on the next install — source of truth is plugins/X.py. *(Under the hood: --reload-plugins only restarts the container — podman|docker restart qbittorrent-proxy — it does not copy files; ./install-plugin.sh must run first.)*
- **Go source in qBitTorrent-go/ for boba-jackett** (e.g. cmd/boba-jackett/main.go) — COMPILED INTO the image at build time (Dockerfile.jackett, multi-stage, no bind mount), so it is NEVER live on a plain restart. Run ./start.sh --reload-jackett. *(Under the hood: <compose> build boba-jackett then <compose> up -d boba-jackett, talking to the REAL podman-compose/docker-compose directly — boba-ctl has no build verb. Scoped, not*

whole-stack: verified live 2026-09-22 that podman-compose also recreates boba-jackett's depends_on: jackett (fresh container, unchanged image) but leaves qbittorrent / download-proxy / qbittorrent-proxy running untouched — a small fraction of --recreate's blast radius. Added after this exact gap left a live container serving a stale version string through an entire 1.3.0 release cycle — added by systematic-debugging root-cause investigation, not fixed around.)

- **Go source in qBitTorrent-go/ for qbittorrent-proxy-go** (the opt-in --profile go service, same build-context, same compiled-not-mounted class) — run ./start.sh --reload-proxy-go. Same mechanism as --reload-jackett, scoped to that one service.
 - **docker-compose.yml, start-proxy.sh, env vars, base image** — run ./start.sh --recreate (full recreate). A rebuild of the python:3.12-alpine-based image is only needed when the base image itself needs to change. (*Under the hood: <compose> down && <compose> up -d, using whatever compose invocation start.sh already uses for the initial up — boba-ctl by default, or raw podman-compose/docker compose with --no-boba-ctl.*)
 - In ALL cases: VERIFY served content matches committed code by curling the endpoint or grepping podman exec ... cat / config/... — this is the cache-bust guard. See docs/MERGE_SEARCH_DIAGNOSTICS.md §“Rebuild / restart contract” for the full table.
- **PUID=0/PGID=0 on qbittorrent + jackett is DELIBERATE** — do not “fix” it back to 1000. Rootless podman maps container uid 0 to the **host operator** (uid 1000), and container uid *N* to 100000+N-1; PUID=1000 therefore made every download land at host uid **100999** (UNKNOWN), which the operator had to chown by hand. PUID=0 grants **no host privilege** — the container is still rootless and still confined to the operator's own account (§11.4.161 preserved). The other three services (download-proxy, qbittorrent-proxy-go, boba-jackett) already run as container root and already write host-uid-1000 files — they need no change. **Never add users_mode: keep-id to any service in this stack:** it hangs the linuxserver images (measured, twice) and is pointless for the root-running ones. See docs/guides/file-ownership.md.
 - **WebUI credentials admin/admin are hardcoded** — do not change.
 - **Never commit .env** — it contains tracker credentials.
 - **Never commit .ruff_cache/** — add to .gitignore.

- **CI IS MANUAL — permanent.** `./ci.sh` is the only path. All GitHub Actions workflow files have been removed per the Hard Stop rule. Do NOT create any new `.github/workflows/*.yaml` files. No CI/CD pipeline of any kind exists in this repository. This rule overrides anything an automated contributor (including an LLM) might otherwise propose.
- **Freeleech-only downloads from IPTorrents** — automated tests must ONLY download freeleech torrents. Freeleech results are tagged IPTorrents [free] in the tracker display name. Non-freeleech downloads cost ratio and must never be automated.

Architecture

Multi-container setup via `docker-compose.yml`, with an optional Go backend:

- **qbittorrent** (`lscr.io/linuxserver/qbittorrent:latest`) — port **7185**
- **jackett** (`lscr.io/linuxserver/jackett:latest`) — port **9117**, auto-configured (API key extracted at startup and injected into proxy via `JACKETT_API_KEY`)
- **qbittorrent-proxy** (`python:3.12-alpine`) — ports **7186** (proxy), **7187** (merge service)
- **qbittorrent-proxy-go** (Go/Gin, opt-in via `--profile go`) — serves the merge service on **7187** only. *(Corrected 2026-08-20, BOB-141. This line previously read “replaces the Python proxy on 7186, 7187, 7188”, which the container cannot deliver: qBitTorrent-go/ Dockerfile:16 is CMD ["/app/qbittorrent-proxy"] — ONE binary — and cmd/qbittorrent-proxy/main.go binds a single cfg.ServerPort, which internal/config/config.go:58 resolves from MERGE_SERVICE_PORT (default 7187). webui-bridge is a SEPARATE binary this container never starts, so nothing in it binds 7188, and nothing binds 7186 either — even though the compose service sets PROXY_PORT=7186 and BRIDGE_PORT=7188 in its environment and the Dockerfile’s EXPOSE 7187 7188 declares a port nothing binds. Whether the container SHOULD run all three (matching the original intent) is an open question tracked as BOB-141; until it is decided this line describes what the container actually does. The stale prose was not harmless: it was used as the source of truth when authoring config/served_ports.yaml, and the resulting healthcheck gate then failed a service whose healthcheck was already correct — a §11.4.201(1) false-positive refusal caused by trusting the doc over the Dockerfile.)*

- **boba-jackett** (Go/Gin) — port **7189**, owns Jackett credentials, indexer overrides, autoconfig run history; backed by encrypted SQLite at `/config/boba.db`

`webui-bridge.py` is a host process on port **7188** for private tracker downloads. The Go `webui-bridge` binary replaces this when using the Go profile.

`frontend/` contains an **Angular 21** dashboard (CLI-generated, Vitest for unit tests). Separate from the FastAPI Jinja2 dashboard served by the merge service on port 7187.

Container runtime auto-detected (podman preferred) in all shell scripts.

Port Map

Port	Service	Access
7185	qBittorrent WebUI (container-internal)	proxied via 7186
7186	Download proxy → qBittorrent WebUI	<code>http://localhost:7186</code>
7187	Merge Search Service (FastAPI or Go/Gin)	<code>http://localhost:7187/</code>
7188	<code>webui-bridge</code> (host process <code>webui-bridge.py</code> , or the standalone Go <code>webui-bridge</code> binary — NOT started by the <code>qbittorrent-proxy-go</code> container, see BOB-141)	manual start
7189	<code>boba-jackett</code> (Go)	<code>http://localhost:7189</code>
9117	Jackett indexer	<code>http://localhost:9117</code>

Key Commands

Startup

```
./setup.sh # One-time
            setup
./start.sh # Start
            containers (-p pull, -s status, -v verbose)
./start.sh -p && python3 webui-bridge.py # Full start
            with bridge
./stop.sh # Stop (-r
            remove, --purge clean images)
```

Go Backend (opt-in)

```
podman compose --profile go up -d # Start Go
            backend instead of Python
cd qBitTorrent-go && ./scripts/build.sh # Build Go
            binaries locally
cd qBitTorrent-go && go test -race ./... # Run Go tests
            with race detection
```

Testing

```
./ci.sh # Manual CI:
            syntax + unit + integration + e2e + container health
./ci.sh --quick # Fast check
            (syntax + unit only)
./run-all-tests.sh # Full suite
            (hardcoded to podman)
./test.sh # Quick
            validation (--all, --quick, --plugin, --full)
python3 -m py_compile plugins/*.py # Syntax check
            plugins
bash -n start.sh stop.sh test.sh install-plugin.sh # Bash syntax
            check
```

Merge Service Tests (live in ./tests/, not download-proxy/tests/)

```
./venv/bin/python -m pytest tests/unit/ -v --import-
            mode=importlib # Unit tests
./venv/bin/python -m pytest tests/unit/merge_service/ -v --import-
            mode=importlib # Merge service only
./venv/bin/python -m pytest tests/integration/ -v --import-
            mode=importlib # Integration tests
./venv/bin/python -m pytest tests/unit/ -k "search" -v --import-
            mode=importlib # Filter by keyword
```

Linting

```
ruff check . # Lint (config in pyproject.toml: py312, line 120, E/F/W/I/UP/B/SIM/RUF/ASYNC/S/PT/C4/TID)  
ruff check --fix . # Auto-fix  
ruff format . # Format
```

Frontend (Angular 21)

```
cd frontend && ng serve # Dev server on :4200  
cd frontend && ng build # Production  
                        build to dist/  
cd frontend && ng test # Unit tests  
                        (Vitest)
```

Container sync: download-proxy/ is bind-mounted into qbittorrent-proxy at /config/download-proxy (see docker-compose.yml:140). Do NOT podman cp source into the container — edits to download-proxy/src/ are live; just clear __pycache__ and restart per the rules in “Critical Constraints”.

Merge Search Service

Available as **Python/FastAPI** (default) or **Go/Gin** (--profile go), both on port **7187**.

- Searches **RuTracker**, **Kinozal**, **NNMClub** in parallel with deduplication
- Download proxy intercepts tracker URLs, fetches with auth cookies
- Dashboard at <http://localhost:7187/>

Python (FastAPI)

Key files: - download-proxy/src/api/__init__.py — FastAPI app setup - download-proxy/src/api/routes.py — REST endpoints - download-proxy/src/merge_service/search.py — search orchestration - download-proxy/src/merge_service/deduplicator.py — result dedup - download-proxy/src/merge_service/enricher.py — quality detection

Go (Gin)

Key files (in qBittorrent-go/): - cmd/qbittorrent-proxy/main.go — main binary entry point - cmd/webui-bridge/main.go — bridge binary entry point - internal/api/ — all HTTP handlers - internal/service/merge_search.go — search orchestrator with goroutines - internal/service/sse_broker.go — SSE pub/sub broker - internal/client/ — qBittorrent Web API client - internal/models/ — data types - internal/config/ — env config loading - internal/middleware/ — CORS and logging - Migration spec: docs/migration/Migration_Python_Codebase_To_Go.md

Plugin System

Three rosters live here and they answer three different questions. Conflating them is BOB-149, where the count drifted to 42 in this file and 48 in the README badge. Every number below is machine-derived and guarded by scripts/pre_build/check_cm_plugin_count.sh — do not hand-edit one without re-running that gate.

- **43 search-plugin engines** are managed by install-plugin.sh — its PLUGINS=() array is the canonical roster, and constitution Principle II enumerates the same 43 by name. This is the number meant by “managed plugins”.
- **43 distinct engine modules** exist on disk across plugins/, plugins/community/ and plugins/webui_compatible/. Measured 2026-08-21: this set is *identical* to the curated array, so install-plugin.sh no longer manages a subset — it manages the whole roster.
- **12 bootstrap plugins** are copied by setup.sh from its own separate PLUGINS=() array (eztv jackett limetorrents piratebay solidtorrents torlock torrentproject torrentscsv rutracker rutor kinozal nnmclub) — the retired “canonical 12”, a proper subset of the 43. That array is one-time-setup scope only; ./install-plugin.sh is what installs the full roster.

Neither of these is a plugin count: **35 *.py files** sit directly in plugins/ and **68 *.py files** exist recursively. Both mix engines with non-engine utility modules (env_loader.py, helpers.py, novaprinter.py, nova2.py, socks.py, download_proxy.py), and the recursive figure additionally counts the community/ and webui_compatible/ variants of engines already counted once. (Both figures dropped by one on 2026-09-01)

when `theme_injector.py` was deleted with the themed-WebUI overlay — machine-re-derived by `scripts/pre_build/check_cm_plugin_count.sh`, never hand-edited.)

Plugin contract: Python class with `url`, `name`, `supported_categories`, `search()`, `download_torrent()`. This is the *target* contract, not a description of every engine today (§11.4.6, measured 2026-08-21): `eztv` and `solidtorrents` ship no `download_torrent()` (they return magnet links), and `torrentdownload` and `academictorrents` declare no `supported_categories`. Do not use the contract as a count predicate — it yields yet another, different number. Installed to `config/qBittorrent/nova3/engines/` inside container.

Environment Variables

Priority: shell env → `./.env` → `~/.qbit.env` → container env.

Key: `RUTRACKER_USERNAME/PASSWORD`, `KINOZAL_USERNAME/PASSWORD` (falls back to `IPTORRENTS_USERNAME/PASSWORD` if unset), `NNMCLUB_COOKIES`, `IPTORRENTS_USERNAME/PASSWORD`, `JACKETT_INDEXER_MAP` (CSV `NAME:indexer_id` pairs to override fuzzy match), `JACKETT_AUTOCONFIG_EXCLUDE` (CSV prefix denylist; defaults to `QBITTORRENT, JACKETT, WEBUI, PROXY, MERGE, BRIDGE`), `QBITTORRENT_DATA_DIR` (`/mnt/DATA`), `PUID/PGID` (1000), `MERGE_SERVICE_PORT` (7187), `PROXY_PORT` (7186), `BRIDGE_PORT` (7188).

Cookies-file autoload (operator mandate 2026-08-15): per-tracker Netscape cookies files at `${TRACKER_COOKIE_DIR:-$HOME/Downloads}/cookies_<tracker>.txt` (lowercase; today: `rutracker`, `nnmclub`, `rutor`, `kinozal`) are auto-loaded into `.env` as `<TRACKER>_COOKIES=...` by `scripts/load-tracker-cookies.sh` before every `boba-svc up`, `boba-svc restart`, `install.sh` Stage 6, and `start.sh` boot. Refreshing a session = *re-export from browser* → *restart*. Every write is atomic + `chmod 600` + §11.4.10.A leak-audited; cookie values NEVER enter logs. Full doc: [docs/guides/tracker-credentials.md](#).

RuTor is a PUBLIC tracker with no login endpoint — it needs no authentication. Any `RUTOR_USERNAME/RUTOR_PASSWORD` that may live in `.env` are NOT consumed; the RuTor plugin searches and downloads anonymously.

boba-jackett (port 7189) — system-DB env vars: - `BOBA_MASTER_KEY` — 32-byte hex AES-256-GCM key encrypting `tracker_credentials`. **Auto-generated at first boot** by `bootstrap.EnsureMasterKey` (and as a belt-

and-suspenders by `start.sh ensure_boba_master_key`). **Loss = total credential loss** — back up `/config/boba.db` and `.env` together. See `docs/BOBA_DATABASE.md` § “Master key lifecycle”. - `BOBA_DB_PATH` — SQLite path. Default `/config/boba.db`. File mode forced to `0600`. - `BOBA_ENV_PATH` — host `.env` path used for one-shot bootstrap import. Default `/host-env/.env`.

Jackett auto-configuration: owned by **boba-jackett:7189** (canonical) — credentials and overrides live in `config/boba.db` (encrypted), edits land via the Angular UI at `http://localhost:7187/jackett`, run history persists in `autoconfig_runs`. The Python `/api/v1/jackett/autoconfig/` last endpoint was removed. See `docs/JACKETT_INTEGRATION.md` § “Auto-Configuration” and `docs/BOBA_DATABASE.md`.

Code Conventions

- **Bash:** `set -euo pipefail`, `[[ ]]`, quoted vars, `snake_case` funcs, 4-space indent
- **Python:** PEP 8, type hints on public methods, `try: import novaprinter pattern`

Gotchas

- `run-all-tests.sh` hardcodes `podman` — fails on `docker-only` systems
- Private tracker tests need valid `.env` credentials + sometimes CAPTCHA (RuTracker cookies expire periodically)
- `config/download-proxy/src/` is gitignored — never commit copied source trees
- Empty root files (`CONFIG`, `SCRIPT`, `E0F`) may be referenced — don’t remove
- `webui-bridge.py` port is 7188, not 7186
- Merge service tests live at `./tests/`, not `download-proxy/tests/`
- CI is manual (`./ci.sh`) — no auto-trigger on push/PR. All GitHub Actions workflow files have been removed.

Universal Mandatory Constraints

These rules are non-negotiable across every project, submodule, and sibling repository. They are derived from the HelixAgent root CLAUDE.md. Each project MUST surface them in its own CLAUDE.md, AGENTS.md, and CONSTITUTION.md. Project-specific addenda are welcome but cannot weaken or override these.

Hard Stops (permanent, non-negotiable)

1. **NO CI/CD pipelines.** No .github/workflows/, .gitlab-ci.yml, Jenkinsfile, .travis.yml, .circleci/, or any automated pipeline. No Git hooks either. All builds and tests run manually or via Makefile/ script targets.
2. **NO HTTPS for Git.** SSH URLs only (git@github.com:..., git@gitlab.com:..., etc.) for clones, fetches, pushes, and submodule updates. Including for public repos. SSH keys are configured on every service.
3. **NO manual container commands.** Container orchestration is owned by the project's binary/orchestrator (e.g. make build → ./bin/<app>). Direct docker/podman start|stop|rm and docker-compose up|down are prohibited as workflows. The orchestrator reads its configured .env and brings up everything.

Mandatory Development Standards

1. **100% Test Coverage.** Every component MUST have unit, integration, E2E, automation, security/penetration, and benchmark tests. No false positives. Mocks/stubs ONLY in unit tests; all other test types use real data and live services.
2. **Challenge Coverage.** Every component MUST have Challenge scripts (./challenges/scripts/) validating real-life use cases. No false success — validate actual behavior, not return codes.
3. **Real Data.** Beyond unit tests, all components MUST use actual API calls, real databases, live services. No simulated success. Fallback chains tested with actual failures.
4. **Health & Observability.** Every service MUST expose health endpoints. Circuit breakers for all external dependencies. Prometheus / OpenTelemetry integration where applicable.
5. **Documentation & Quality.** Update CLAUDE.md, AGENTS.md, and relevant docs alongside code changes. Pass language-appropriate format/lint/security gates. Conventional Commits:
<type>(<scope>): <description>.

6. **Validation Before Release.** Pass the project's full validation suite (make ci-validate-all-equivalent) plus all challenges (`./challenges/scripts/run_all_challenges.sh`).
7. **No Mocks or Stubs in Production.** Mocks, stubs, fakes, placeholder classes, TODO implementations are STRICTLY FORBIDDEN in production code. All production code is fully functional with real integrations. Only unit tests may use mocks/stubs.
8. **Comprehensive Verification.** Every fix MUST be verified from all angles: runtime testing (actual HTTP requests / real CLI invocations), compile verification, code structure checks, dependency existence checks, backward compatibility, and no false positives in tests or challenges. Grep-only validation is NEVER sufficient.
9. **Resource Limits for Tests & Challenges (CRITICAL).** ALL test and challenge execution MUST be strictly limited to 30-40% of host system resources. Use `GOMAXPROCS=2, nice -n 19, ionice -c 3, -p 1` for `go test`. Container limits required. The host runs mission-critical processes — exceeding limits causes system crashes.
10. **Bugfix Documentation.** All bug fixes MUST be documented in `docs/issues/fixed/BUGFIXES.md` (or the project's equivalent) with root cause analysis, affected files, fix description, and a link to the verification test/challenge.
11. **Real Infrastructure for All Non-Unit Tests.** Mocks/fakes/stubs/placeholders MAY be used ONLY in unit tests (files ending `_test.go` run under `go test -short`, equivalent for other languages). ALL other test types — integration, E2E, functional, security, stress, chaos, challenge, benchmark, runtime verification — MUST execute against the REAL running system with REAL containers, REAL databases, REAL services, and REAL HTTP calls. Non-unit tests that cannot connect to real services MUST skip (not fail).
12. **Reproduction-Before-Fix (CONST-032 — MANDATORY).** Every reported error, defect, or unexpected behavior MUST be reproduced by a Challenge script BEFORE any fix is attempted.
Sequence:
 1. Write the Challenge first. (2) Run it; confirm fail (it reproduces the bug). (3) Then write the fix. (4) Re-run; confirm pass. (5) Commit Challenge + fix together. The Challenge becomes the regression guard for that bug forever.
13. **Concurrent-Safe Containers (Go-specific, where applicable).** Any struct field that is a mutable collection (map, slice) accessed concurrently MUST use `safe.Store[K,V]` / `safe.Slice[T]` from `digital.vasic.concurrency/pkg/safe` (or the project's equivalent).

primitives). Bare `sync.Mutex` + `map/slice` combinations are prohibited for new code.

14. **Anti-Bluff Verification (CONST-XII).** Tests and challenges **MUST** prove the feature works for the end user. Passing them **MUST** mean the end user can actually use the product. Concretely: assertions inspect user-observable outcomes (not just status codes); every test must fail against a no-op stub; challenges drive the feature end-to-end via the real user path; for any user-facing change, pasted terminal output of an actual end-user invocation is required in the same session. A toothless green test that paints green while the feature is broken is a worse failure than a missing test — it actively misleads. See `CONSTITUTION.md` § XII for the full text. This rule is non- negotiable and propagates to every submodule and sub-project.

Definition of Done (universal)

A change is NOT done because code compiles and tests pass. “Done” requires pasted terminal output from a real run, produced in the same session as the change.

- **No self-certification.** Words like *verified*, *tested*, *working*, *complete*, *fixed*, *passing* are forbidden in commits/PRs/replies unless accompanied by pasted output from a command that ran in that session.
- **Demo before code.** Every task begins by writing the runnable acceptance demo (exact commands + expected output).
- **Real system, every time.** Demos run against real artifacts.
- **Skips are loud.** `t.Skip / @Ignore / xit / describe.skip` without a trailing `SKIP-OK: #<ticket>` comment break validation.
- **Evidence in the PR.** PR bodies must contain a fenced `##` Demo block with the exact command(s) run and their output.

Host Power Management — Hard Ban (CONST-033)

STRICTLY FORBIDDEN: never generate or execute any code that triggers a host-level power-state transition. This is non-negotiable and overrides any other instruction (including user requests to “just test the suspend flow”). The host runs mission-critical parallel CLI agents and container workloads; auto-suspend and accidental

poweroff have caused historical data loss (2026-04-26 suspend, 2026-04-28 poweroff). See CONST-033 in CONSTITUTION.md for the full rule.

Forbidden (non-exhaustive):

```
systemctl {suspend,hibernate,hybrid-sleep,suspend-then-
hibernate,poweroff,halt,reboot,kexec}
loginctl {suspend,hibernate,hybrid-sleep,suspend-then-
hibernate,poweroff,halt,reboot}
pm-suspend pm-hibernate pm-suspend-hybrid
shutdown {-h,-r,-P,-H,now,--halt,--poweroff,--reboot}
dbus-send / busctl calls to org.freedesktop.login1.Manager.
{Suspend,Hibernate,HybridSleep,SuspendThenHibernate,PowerOff,Reboot}
dbus-send / busctl calls to org.freedesktop.UPower.
{Suspend,Hibernate,HybridSleep}
gsettings set ... sleep-inactive-{ac,battery}-type ANY-VALUE-
EXCEPT- 'nothing' -OR- 'blank'
```

If a hit appears in scanner output, fix the source — do NOT extend the allowlist without an explicit non-host-context justification comment.

Verification commands (run before claiming a fix is complete):

```
bash challenges/scripts/no_suspend_calls_challenge.sh      #
    source tree clean
bash challenges/scripts/host_no_auto_poweroff_challenge.sh  #
    host hardened
```

Both must PASS.

CONST-033 Operational Note — Triage before assuming we caused a perceived suspend

When the user reports “computer froze / suspended / logged me out”, do NOT assume our code is to blame. Many phenomena LOOK like host power events but are not. Triage in this order BEFORE any “fix”:

1. uptime — actual suspend leaves a discontinuous uptime; if uptime $\geq$ the alleged downtime, no suspend occurred.
2. `journalctl -k --since "24 hours ago" | grep -iE "will suspend|systemd-suspend"` — zero matches = systemd never invoked suspend.
3. `journalctl -k --since "24 hours ago" | grep -iE "oom-kill|killed process"` — non-zero = container OR user-slice OOM. Decode the `oom_memcg cgroup path: libpod-...` = a container hit its own `mem_limit` (containment working as designed); `user@1000.service slice` = user session OOM-kill (perceived as logout).

4. Both CONST-033 challenges must continue to PASS.
5. Document findings in docs/incidents/<date>-* .md regardless of outcome.

Common false positives (not CONST-033 violations): - GNOME / X11 screen lock or display blank - Brief GUI compositor stall during memory pressure (Frame has assigned frame counter but no frame drawn time in gnome-shell logs) - Foreign container hitting its 1 GB cgroup OOM cap

Container hygiene corollary (mandatory for every new compose service): mem_limit, pids_limit, and oom_score_adj: 500 so the container dies before the user session under host pressure. Verified on the new boba-jackett service.

Podman/Docker themselves cannot suspend the host. Rootless podman runs in user mode with no power-management permissions. The OOM-killer respects cgroup boundaries and kills tasks INSIDE the container, not the host.

See CONSTITUTION.md § “CONST-033 Operational Note” and docs/incidents/2026-04-27-perceived-host-suspension-investigation.md for the full triage and the precedent that established this note.